

PART 1: BASIC CONCEPTS OF NETWORKING AND TELECOMMUNICATION

Chapter 1: Computer Network and Telecommunication

1. Computer Network

Definition of Computer Network

A computer network is a collection of two or more computers and other devices connected together through communication media in order to share data, information, hardware resources, software resources, and services.

Objectives of Computer Network

The main objectives of a computer network are:

1. To share data and information between users.
2. To share hardware devices such as printers and scanners.
3. To enable communication among users.
4. To provide internet access to multiple devices.
5. To improve efficiency and reduce operational cost.

Examples of Computer Network

- Computers connected in a school computer laboratory.
- ATM networks used by banks.
- Office networks used for sharing files and printers.
- The Internet.

Advantages of Computer Network

1. Resource Sharing

Network allows users to share hardware devices such as printers, scanners, and storage devices.

2. Fast Communication

Users can communicate quickly using email, messaging, and video conferencing.

3. Data Sharing

Files and information can be easily shared among users.

4. Centralized Management

Data and applications can be managed from a central location.

5. Cost Reduction

Sharing resources reduces the overall cost of equipment and maintenance.

Disadvantages of Computer Network

1. Security Risks

Unauthorized users and hackers may access network resources.

2. Virus and Malware Spread

Viruses can spread quickly from one computer to another through a network.

3. High Setup Cost

Installing network devices and cables can be expensive.

4. Network Failure

If the network fails, communication and resource sharing stop.

2. Computer Telecommunication

Definition of Computer Telecommunication

Computer telecommunication is the process of transmitting data, information, text, audio, and video between computers and communication devices over a distance through electronic communication systems.

Examples of Computer Telecommunication

- Sending emails
- Video conferencing
- Online chatting
- Mobile communication
- Internet browsing

Importance of Computer Telecommunication

1. Global Communication

People from different countries can communicate instantly.

2. Online Education

Students can attend classes remotely through the internet.

3. Business Communication

Organizations can communicate with clients and employees easily.

4. Information Sharing

Information can be shared quickly across the world.

3. Broadband

Traditional internet connections were slow and could not support modern online activities such as video streaming and online gaming. Broadband technology provides high-speed internet connections capable of transferring large amounts of data continuously.

Definition of Broadband

Broadband is a high-speed internet connection technology that allows continuous transmission of large amounts of data over communication media.

Features of Broadband

1. Provides high-speed internet access.
2. Supports continuous connection.
3. Allows multimedia services such as video streaming and online gaming.
4. Supports multiple users simultaneously.
5. Provides better communication quality.

Types of Broadband

3.1 Fiber Optic Internet

Fiber optic internet is a broadband technology that transmits data using light signals through optical fiber cables.

3.2 DSL (Digital Subscriber Line)

Definition

DSL is a broadband technology that provides internet access using traditional telephone lines.

3.3 ADSL (Asymmetric Digital Subscriber Line)

Definition

ADSL is a type of DSL technology in which download speed is higher than upload speed.

3.4 Cable Internet

Definition

Cable internet is a broadband technology that uses television cable lines to provide internet access.

3.5 Satellite Internet

Definition

Satellite internet is a communication technology that provides internet access through satellites located in space.

3.6 WiFi Connection

Definition

WiFi is a wireless networking technology that allows devices to connect to the internet without using physical cables.

3.7 Mobile Broadband

Definition

Mobile broadband is wireless internet access provided through mobile network technologies.

Types of Mobile Broadband

1. 3G
2. 4G
3. 5G

4. Bandwidth

Definition of Bandwidth

Bandwidth is the maximum amount of data that can be transmitted through a communication channel in a given amount of time.

Units of Bandwidth

Bandwidth is measured in:

- bps (bits per second)
- Kbps (Kilobits per second)
- Mbps (Megabits per second)
- Gbps (Gigabits per second)

Example

A network connection with 100 Mbps bandwidth can theoretically transfer 100 million bits of data per second.

Importance of Bandwidth

1. Determines network speed.
2. Supports large file transfer.
3. Improves video streaming quality.
4. Allows multiple users to use the internet efficiently.

5. Throughput

Definition

Throughput is the actual amount of data successfully transmitted through a network in a given period of time.

Explanation

Bandwidth shows the maximum capacity of a network, whereas throughput shows the real performance of the network.

Example

If a network has 100 Mbps bandwidth but transfers only 70 Mbps due to congestion, then the throughput is 70 Mbps.

Factors Affecting Throughput

1. Network congestion
2. Signal interference
3. Hardware limitations
4. Distance between devices
5. Network traffic

Difference Between Bandwidth and Throughput

Basis	Bandwidth	Throughput
Definition	Maximum data transfer capacity	Actual transferred data
Nature	Theoretical	Practical
Speed	Usually higher	Usually lower
Affected by	Network design	Network condition
Measurement	Mbps, Gbps	Mbps, Gbps

6. Data Packets

Introduction

When data is transmitted over a network, it is not sent as a single large unit. Instead, it is broken into smaller parts for efficient transmission. These small units are called data packets.

Definition of Data Packet

A data packet is a small unit of data that is transmitted over a network, which contains actual data along with control information such as source address, destination address, and sequencing information.

Structure of a Data Packet

A data packet generally consists of the following parts:

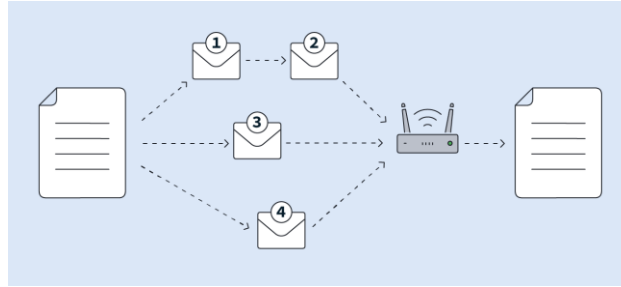
1. **Header** – It contains the source address, destination address, and control information.
2. **Data (Payload)** – It contains the actual information being transmitted.
3. **Trailer** – It contains error detection information to ensure data integrity.

Importance of Data Packets

1. Data packets allow efficient transmission of data over networks.
2. They help in error detection and correction.
3. They ensure data is delivered in the correct order.
4. They reduce network congestion by dividing large data into smaller parts.

Example

When a large file such as a video is sent over the internet, it is divided into multiple packets. These packets may travel through different routes and are reassembled at the destination in the correct order.



7. Frequency

Introduction

In communication systems, signals are transmitted using electromagnetic waves. These signals vary in their rate of oscillation, which is known as frequency.

Definition of Frequency

Frequency is the number of complete cycles or waves of a signal transmitted per second in a communication system.

Unit of Frequency

The unit of frequency is Hertz (Hz).

Types of Frequency

1. High Frequency

High frequency refers to signals that have a large number of cycles per second.

Characteristics:

- Carries more data.
- Suitable for fast communication.
- Covers shorter distances.

Examples:

- WiFi networks
- 4G and 5G networks

2. Low Frequency

Low frequency refers to signals that have fewer cycles per second.

Characteristics:

- Carries less data.
- Travels longer distances.
- More stable in long-distance communication.

Examples:

- Radio broadcasting
- AM radio signals

Example: Radio broadcasting.

8. Modes of Data Transmission (Communication Modes)

Introduction

Data transmission between devices can occur in different directions depending on the communication system. These modes are called communication modes.

1. Simplex Mode

Definition

Simplex mode is a communication mode in which data flows only in one direction from sender to receiver.

Explanation

In simplex communication, the receiver cannot send data back to the sender.

Example

- Television broadcasting
- Keyboard to computer communication

2. Duplex Mode

Definition

Duplex mode is a communication mode in which data can flow in both directions between sender and receiver.

i. Half Duplex Mode

Definition

Half duplex mode is a communication mode in which data can flow in both directions, but not at the same time.

Example

- Walkie-talkie communication

Explanation

When one person is speaking, the other must wait to respond.

ii. Full Duplex Mode

Definition

Full duplex mode is a communication mode in which data can flow in both directions simultaneously.

Example

- Telephone conversation

Explanation

Both users can speak and listen at the same time.

Difference Between Simplex, Half Duplex, and Full Duplex

Mode	Direction	Communication
Simplex	One-way	Only sender to receiver
Half Duplex	Two-way	One at a time
Full Duplex	Two-way	Simultaneous

9. Difference between 3G, 4G and 5G

Feature	3G	4G	5G
Introduced	2001	2009	2018
Technology	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	CDMA	CDMA	OFDM, BDMA
Switching type	Packet switching, except for air interference	Packet switching	Packet switching

Internet service	Broadband	Ultra-broadband	Wireless World Wide Web
Bandwidth	25 MHz	100 MHz	30–300 GHz
Speed	up to 2 Mbps	up to 100 Mbps	up to 10 Gbps
Latency	100–500 ms	30–50 ms	as low as 1 ms
Data capacity	Limited	Much higher than 3G	Extremely high, supports IoT and massive connections
Best for	Web browsing, emails, basic apps	Streaming, video calls, gaming	Smart cities, AR/VR, real-time applications
Cost for users	Cheaper, but slow	Standard for most users	Generally higher costs for early adopters

9. Communication Channel

Definition

A communication channel is a transmission medium through which data is sent from a sender to a receiver in a network.

Types of Communication Channel

1. Wired (Guided) Communication Media
2. Wireless (Unguided) Communication Media

10. Wired (Guided) Communication Media

Definition

Wired communication media are those transmission media in which data is transmitted through physical cables or wires.

Features of Wired Media

- High speed communication
- More secure than wireless
- Less interference
- Requires physical installation

10.1 Twisted Pair Cable

Definition

A twisted pair cable is a type of communication cable that consists of two insulated copper wires twisted together to reduce electromagnetic interference.

Features

- Low cost
- Easy installation
- Suitable for short distance communication
- Used in LAN networks

Types of Twisted Pair Cable

i. Unshielded Twisted Pair (UTP)

Definition

UTP is a type of twisted pair cable that does not have any additional shielding for protection against interference.

Uses

- Telephone networks
- LAN connections in homes and offices

Characteristics

- Low cost
- Easy to install
- More interference compared to STP

ii. Shielded Twisted Pair (STP)

Definition

STP is a type of twisted pair cable that includes an additional shielding layer to reduce electromagnetic interference.

Uses

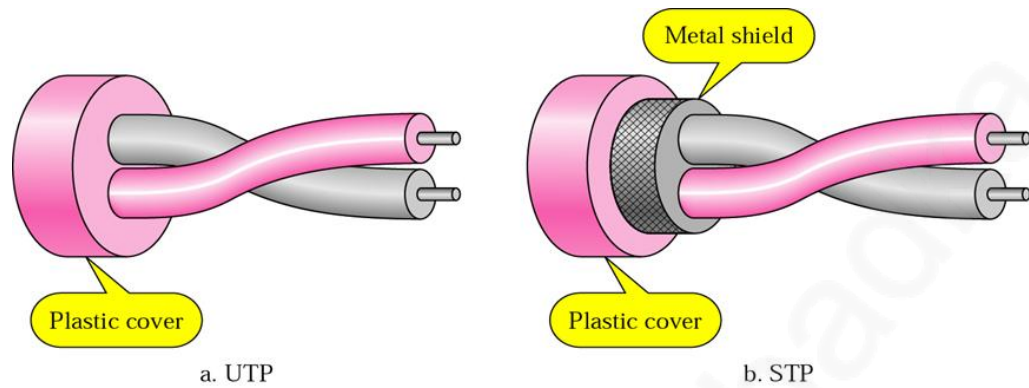
- Industrial networks
- High-noise environments

Characteristics

- Less interference

- More expensive than UTP
- Better performance

Parameter	UTP	STP
Full Form	UTP stands for Unshielded Twisted Pair.	STP stands for Shielded Twisted Pair.
Grounding Cable	In UTP grounding cable is not necessary.	While in STP grounding cable is required.
Data Rate	Data rate in UTP is slow compared to STP.	Data rate in STP is high.
Cost	The cost of UTP is less.	While STP is costlier than UTP.
Maintenance	Less maintenance needed.	Much more maintenance is needed.
Noise	Noise is high compared to STP.	Noise is less.
Generation of crosstalk	The generation of crosstalk is also high compared to STP.	Generation of crosstalk is also less.
Attenuation	Attenuation is high in comparison to STP.	<u>Attenuation</u> is low.
Speed	About 10 to up to 1000 Mbps.	About 10 to up to 100 Mbps.
Installation	Easy and inexpensive	Expensive than UTP and difficult to install.
Max nodes	1024	270
used for	Data transmission within short distance such as for home and office networks.	Connecting organizations over a long distance.
Electromagnetic Interference(EMI)	Electromagnetic interference is more in Unshielded Twisted Pair cable compared to Shielded Twisted Pair cable.	<u>Shielded Twisted Pair cable</u> reduces Electromagnetic interference because of the protective sheath.
Catagories	UTP cables categories as specified by EIA- Category-1, Category-2, Category-3, Category-4, Category-5, Category-5e,Category-6, Category-6a and Category-7.	Shielded cables have commonly these configurations- Foil Shielded and Braid Shielded.
Usage	Used for Telephone wiring, <u>Local Area Networks (LAN)</u> and more.	Used in frigid temperature,employed under highg radiation etc.



10.2 CAT6 Cable

Definition

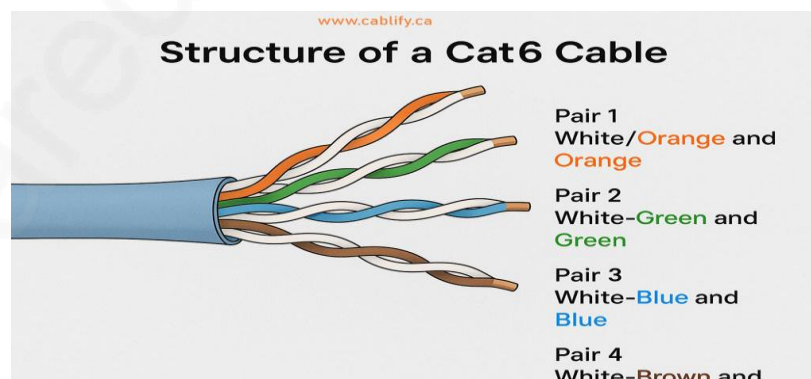
CAT6 cable is an advanced type of twisted pair cable used for high-speed networking.

Features

- High data transfer speed
- Reduced crosstalk
- Supports Gigabit Ethernet

Uses

- Office networks
- Data centers
- High-speed LAN connections



10.3 Optical Fiber Cable

Definition

Optical fiber cable is a high-speed communication cable that transmits data in the form of light signals through glass or plastic fibers.

Structure

1. Core – Carries light signals
2. Cladding – Reflects light back into the core
3. Protective layer – Protects the fiber

Features

- Very high speed
- Long distance communication
- Immune to electromagnetic interference

Advantages

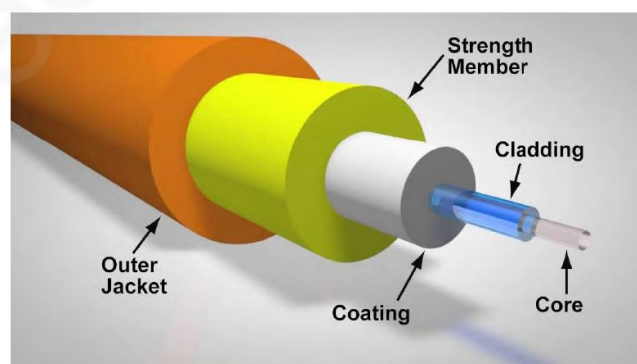
- High bandwidth
- Fast data transmission
- Secure communication

Disadvantages

- Expensive
- Difficult to install and repair

Uses

- Internet backbone
- Telecommunication systems
- Cable television



11. Wireless (Unguided) Communication Media

Definition

Wireless communication media is a type of communication medium in which data is transmitted from one device to another without using any physical conductor such as cables or wires.

Features

1. It does not require physical cables for data transmission.
2. It allows mobility and flexibility for users.
3. It is easy to install and maintain.
4. It is affected by interference and environmental conditions.
5. It is widely used in modern communication systems.

11.1 WiFi (Wireless Fidelity)

Definition

WiFi is a wireless networking technology that allows devices to connect to the internet or local network using radio waves without physical cables.

Features of WiFi

1. It provides wireless internet access.
2. It supports multiple devices at the same time.
3. It has a limited coverage area.
4. It is commonly used in homes, schools, and offices.

Uses of WiFi

- Internet access in homes and institutions
- Connecting laptops, mobiles, and smart devices
- Wireless communication in offices

11.2 Bluetooth

Definition

Bluetooth is a short-range wireless communication technology used for exchanging data between devices over a short distance.

Features of Bluetooth

1. It has a short communication range (usually up to 10 meters).
2. It consumes low power.
3. It is used for device-to-device communication.

4. It does not require internet connection.

Uses of Bluetooth

- Connecting wireless headphones and speakers
- Transferring files between mobile devices
- Connecting keyboards and mouse wirelessly

11.3 RFID (Radio Frequency Identification)

Definition

RFID is a wireless communication technology that uses radio waves to automatically identify and track objects using tags and readers.

Features of RFID

1. It does not require physical contact.
2. It can read multiple tags simultaneously.
3. It is fast and efficient.
4. It is widely used for tracking and identification.

Uses of RFID

- Student ID cards in schools
- Inventory management in shops
- Security access systems

11.4 Satellite Communication

Definition

Satellite communication is a wireless communication system in which satellites orbiting the earth are used to transmit signals between sender and receiver over long distances.

Components of Satellite Communication

i. Space Segment

The space segment consists of satellites placed in space that receive, amplify, and retransmit signals back to earth.

ii. Ground Segment

The ground segment consists of earth-based stations that send and receive signals from satellites.

Features of Satellite Communication

1. It covers a very large geographical area.
2. It supports long-distance communication.
3. It is used in remote and inaccessible areas.
4. It is expensive to install and maintain.

Uses of Satellite Communication

- Television broadcasting
- Weather forecasting
- GPS navigation systems
- Internet services in remote areas

Features	Wired Connection	Wireless Connection
Connection Medium	Physical cables	Radio waves (Wi-Fi signals)
Speed	Typically faster and more stable (up to 1 Gbps or higher)	Generally slower and variable, up to around 600 Mbps to several Gbps depending on standard (e.g., Wi-Fi 6)
Reliability	Very reliable with minimal interference	Prone to interference from obstacles, devices, and distance
Security	More secure due to physical access requirement	More vulnerable to hacking but improved with modern encryption protocols
Mobility	Limited by cable length; device must be tethered	Allows device mobility within signal range
Installation	Involves running cables and management	Easier and quicker to deploy without cables
Latency	Lower latency, better for real-time applications	Higher latency, which may affect gaming or video calls
Scalability	Requires physical cabling for expansion	Easily scalable by adding access points or routers
Use Cases	Suitable for high-performance needs like gaming, video conferencing, secure corporate environments	Ideal for flexible environments, public spaces, homes, and mobile devices

12. Connector

Definition of Connector

A connector is a hardware device used to physically connect network cables to computers, routers, switches, and other networking devices for data transmission.

Importance of Connector

1. It ensures proper connection between network devices.
2. It allows stable data transmission.
3. It prevents signal loss and loose connections.

12.1 RJ45 Connector

Definition

RJ45 (Registered Jack 45) is a standard connector used for connecting Ethernet cables in computer networking.

Features of RJ45 Connector

1. It has 8 pins for data transmission.
2. It is commonly used in LAN (Local Area Network).
3. It provides reliable and stable connection.
4. It is used with twisted pair cables (UTP/STP).

Uses of RJ45 Connector

- Connecting computers to switches
- Connecting routers to modems
- Building LAN networks



12.2 Media Converter

Definition

A media converter is a networking device that converts one type of transmission media into another type of media for communication.

Features of Media Converter

1. It converts electrical signals into optical signals or vice versa.
2. It enables communication between different types of networks.
3. It extends network distance.

Uses of Media Converter

- Connecting fiber optic networks with copper networks
- Extending LAN networks over long distances
- Improving network flexibility

Example

A media converter can convert data from a copper cable (Ethernet) into light signals for transmission through optical fiber cables.

Summary (Quick Revision Points)

Wireless Media includes:

- WiFi
- Bluetooth
- RFID
- Satellite Communication

Connector Devices include:

- RJ45 Connector
- Media Converter

13. Network Components (Networking Devices)

Introduction

In a computer network, different devices are required to transmit, receive, and manage data efficiently. These devices are known as network components or networking devices. They play an important role in establishing communication between computers in a network.

Definition of Network Components

Network components are the hardware and software elements that are used to establish, manage, and control communication between devices in a computer network.

Types of Network Components

Network components are mainly classified into:

1. Software Components

2. Hardware Components

13.1 Software Components

Definition

Software components are the programs and applications that control and manage the operation of a computer network.

Examples

- Network Operating System (NOS)
- Network management software
- Protocol software

Functions

1. Managing network communication
2. Controlling data flow
3. Providing security and access control
4. Managing network resources

13.2 Hardware Components

Definition

Hardware components are physical devices used in a computer network to transmit, receive, and manage data.

1. Repeater

Definition

A repeater is a networking device that receives a weak signal, regenerates it, and retransmits it to extend the distance of the network.

Function

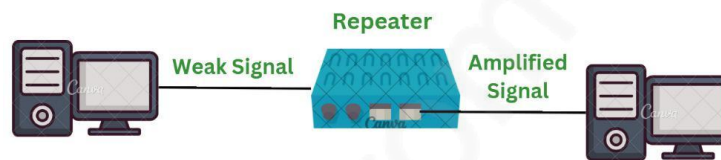
- It strengthens weak signals.

Features

- Works at physical layer
- Extends network range
- Does not filter data

Use

- Used in long-distance communication networks



2. Hub

Definition

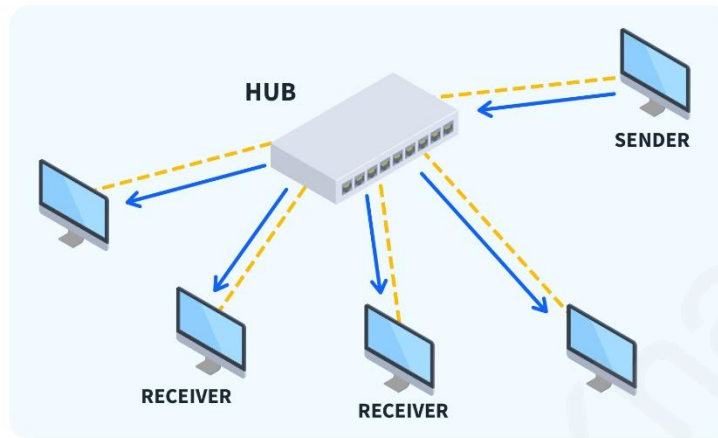
A hub is a networking device that connects multiple computers in a network and broadcasts data to all connected devices.

Features

- Operates at physical layer
- Sends data to all devices
- Simple and inexpensive

Disadvantages

- Low security
- High network traffic



3. Switch

Definition

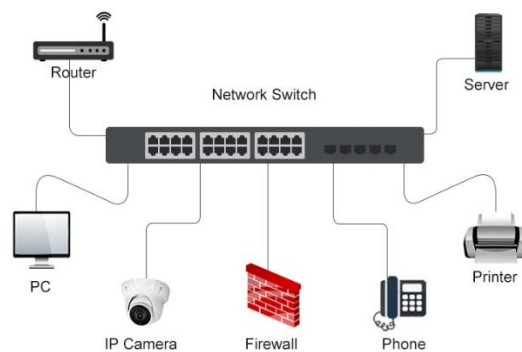
A switch is an intelligent networking device that connects multiple devices and forwards data only to the intended receiver.

Features

- Works at data link layer
- Reduces network traffic
- Provides better security than hub

Advantages

- Efficient data transmission
- Less collision
- Faster communication



4. Bridge

Definition

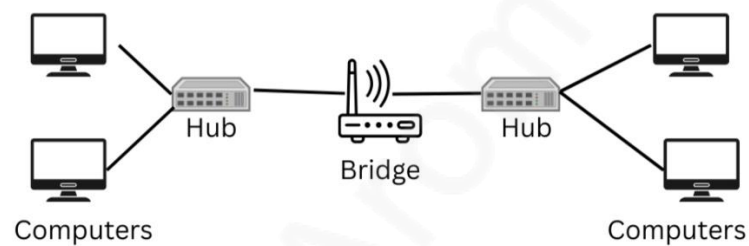
A bridge is a networking device that connects two similar LAN networks and filters data between them.

Features

- Works at data link layer
- Divides network into segments
- Reduces network traffic

Use

- Connecting two LANs in a building



5. Router

Definition

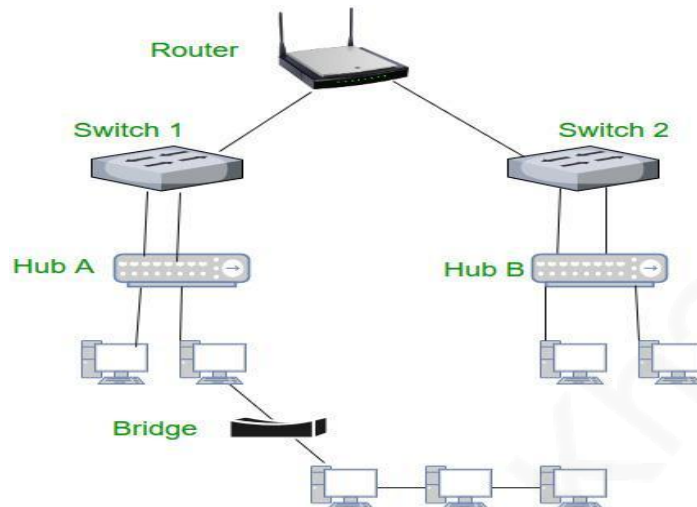
A router is a networking device that connects different networks and directs data packets between them using IP addresses.

Features

- Works at network layer
- Connects LAN to internet
- Provides routing of data

Functions

1. Determines best path for data
2. Connects different networks
3. Provides network security



6. Difference between Hub and Switch

Hub	Switch
<u>Hub</u> is operated on Physical layer of OSI model .	While <u>switch</u> is operated on Data link layer of OSI Model .
Hub is a broadcast type transmission.	While switch is a Unicast, multicast and broadcast type transmission.
Hub have 4/12 ports.	While switch can have 24 to 48 ports.
In hub, there is only one collision domain.	While in switch, different ports have own collision domain.
Hub is a half duplex transmission mode.	While switch is a full duplex transmission mode.
In hub, Packet filtering is not provided.	While in switch, Packet filtering is provided.
Hub cannot be used as a repeater.	While switch can be used as a repeater.
Hub is not an intelligent device that sends message to all ports hence it is comparatively inexpensive.	While switch is an intelligent device that sends message to selected destination so it is expensive.
Hub is simply old type of device and is not generally used.	While switch is very sophisticated device and widely used.
Hacking of systems attached to hub is complex.	Hacking of systems attached to switch is little easy.
Speed of original hub 10Mbps and modern internet hub is 100Mbps.	Maximum speed is 10Mbps to 100Mbps.
Hubs are used in LANs.	Switch is used in LANs.
Cheaper as compared to switch.	Expensive as compared to HUB.

7. Difference between Router and Switch

Router	Switch
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The main objective of router is to connect various networks simultaneously.	While the main objective of switch is to connect various devices simultaneously.
It works in network layer .	While it works in data link layer .
Router is used by LAN as well as MAN .	While switch is used by only LAN.
Through the router, data is sent in the form of packets.	While through switch data is sent in the form of frame.
There is less collision taking place in the router.	While there is no collision taking place in full duplex switch.
Router is compatible with NAT .	While it is not compatible with NAT.
Router is a relatively much more expensive device than switch.	Switch is an expensive device than hub . but cheaper than router.
maximum speed for wireless is 1-10 Mbps and maximum speed for wired connections is 100 Mbps.	Maximum speed is 10Mbps to 100Mbps.
Router needs at least two networks to connect.	Switch needs at least single network is to connect.
The types of routing are: Adaptive and Non-adaptive routing .	The types of switching are: Circuit , Packet , and Message Switching.

14. Network Topology

Definition of Network Topology

Network topology is the physical or logical arrangement of computers, cables, and devices in a computer network.

Types of Network Topology

1. Bus Topology

Definition

Bus topology is a type of network in which all devices are connected to a single main cable called a backbone.

Features

- Single communication line
- Easy to install
- Cost effective

Advantages

- Requires less cable
- Easy to set up

- Suitable for small networks

Disadvantages

- Failure of backbone affects entire network
- Difficult to troubleshoot
- Performance decreases with more devices

2. Star Topology

Definition

Star topology is a network in which all devices are connected to a central device such as a switch or hub.

Features

- Centralized control
- Easy troubleshooting
- Widely used topology

Advantages

- Failure of one device does not affect others
- Easy to manage
- High performance

Disadvantages

- Failure of central device stops network
- Requires more cable

3. Ring Topology

Definition

Ring topology is a network in which each device is connected to two other devices forming a circular network.

Features

- Data flows in circular direction
- No central device
- Token-based communication (in some systems)

Advantages

- Equal access to all devices
- No data collision

Disadvantages

- Failure of one device affects whole network
- Difficult to troubleshoot

4. Hybrid Topology

Definition

Hybrid topology is a combination of two or more different types of network topologies.

Features

- Flexible design
- Scalable network
- Used in large organizations

Advantages

- Highly reliable
- Supports large networks
- Flexible structure

Disadvantages

- Expensive
- Complex design and maintenance

15. Network Types (Based on Coverage Area)

Introduction

Computer networks can be classified according to the geographical area they cover. The size of the network determines its speed, purpose, and usage. Based on coverage, networks are mainly divided into PAN, LAN, MAN, and WAN.

15.1 Personal Area Network (PAN)

Definition

A Personal Area Network (PAN) is a type of computer network that connects electronic devices within a very small range around a single person.

Features of PAN

1. It covers a very small area, usually within 1 to 10 meters.
2. It is used for personal communication and device connectivity.
3. It uses wireless technologies such as Bluetooth or USB connections.
4. It is simple and easy to set up.

Examples

- Bluetooth connection between mobile phone and headphones
- Connecting a smartwatch to a mobile phone

15.2 Local Area Network (LAN)

Definition

A Local Area Network (LAN) is a type of network that connects computers and devices within a small geographical area such as a room, building, or campus.

Features of LAN

1. It covers a limited area like a school or office.
2. It provides high-speed data communication.
3. It is privately owned and managed.
4. It uses Ethernet cables or WiFi for connection.

Examples

- Computer lab in a school
- Office network in a company
- Home WiFi network

Advantages

- High speed communication
- Easy file and resource sharing
- Low cost maintenance

Disadvantages

- Limited coverage area
- Security risks if not protected properly

5.3 Metropolitan Area Network (MAN)

Definition

A Metropolitan Area Network (MAN) is a network that covers a large geographical area such as a city or a town.

Features of MAN

1. It covers a city or large campus.
2. It connects multiple LANs together.
3. It provides high-speed internet services.
4. It is larger than LAN but smaller than WAN.

Examples

- City-wide internet network
- Cable television network in a city
- University campus network spread across multiple buildings

Advantages

- Covers large area
- High speed communication between LANs
- Efficient for city services

Disadvantages

- Expensive to install and maintain
- Requires advanced infrastructure

15.4 Wide Area Network (WAN)

Definition

A Wide Area Network (WAN) is a type of network that covers a very large geographical area such as a country or the whole world.

Features of WAN

1. It covers very large geographical distances.
2. It connects multiple LANs and MANs.
3. It uses satellite links, fiber optics, and telephone lines.
4. It is the largest type of computer network.

Examples

- The Internet
- Banking network systems
- Airline reservation systems

Advantages

- Global communication
- Supports large organizations
- Connects remote areas

Disadvantages

- Very expensive
- Complex setup
- Slower than LAN in some cases

Difference Between PAN, LAN, MAN, and WAN

Feature	PAN	LAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Very small (1–10m)	Building or campus	City	Country/World
Speed	Low to moderate	High	Moderate to high	Variable
Ownership	Personal	Private organization	Government/ISP	Multiple organizations
Example	Bluetooth	School lab	City network	Internet

16. Network Architecture

Definition of Network Architecture

Network architecture is the design and structure of a computer network that defines how devices are connected and how data is transmitted between them.

Types of Network Architecture

16.1 Client-Server Architecture

Definition

Client-Server architecture is a network model in which one central computer called a server provides services, data, and resources to multiple computers called clients.

Working Principle

In this architecture, the client sends a request to the server, and the server processes the request and sends back the response.

Features of Client-Server Architecture

1. Centralized control is maintained by the server.
2. Data and resources are stored on the server.
3. Clients depend on the server for services.
4. It provides high security and data management.

Examples

- Web servers providing websites
- School management systems
- Online banking systems

Advantages

- High security
- Centralized data management
- Easy backup and recovery
- Efficient resource management

Disadvantages

- Expensive setup
- Server failure affects entire network
- Requires skilled maintenance

16.2 Peer-to-Peer (P2P) Architecture

Definition

Peer-to-Peer architecture is a network model in which all computers are equal and can act as both client and server to share resources directly.

Working Principle

In P2P architecture, each computer can request and provide services without a central server.

Features of Peer-to-Peer Architecture

1. No central server is required.

2. All computers have equal roles.
3. Data is shared directly between devices.
4. Simple and easy to install.

Examples

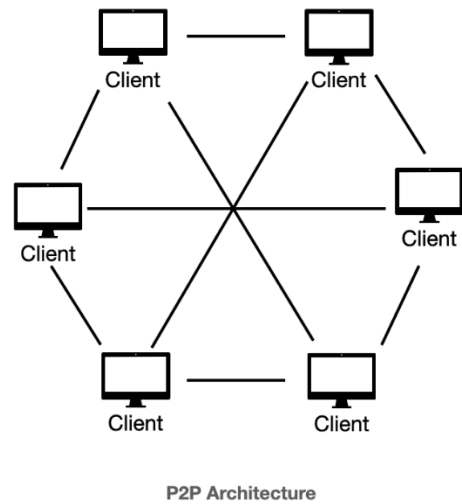
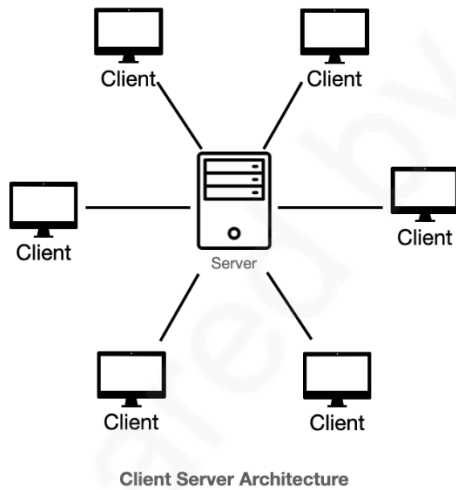
- File sharing networks
- Small office networks
- Bluetooth file sharing

Advantages

- Low cost
- Easy to set up
- No dedicated server required

Disadvantages

- Less security
- Difficult to manage in large networks
- No centralized control



Difference Between Client-Server and Peer-to-Peer Architecture

Feature	Client-Server	Peer-to-Peer
Structure	Central server exists	No central server
Control	Centralized	Distributed
Security	High	Low

Cost	Expensive	Cheap
Scalability	High	Limited
Example	Web services	File sharing

4. Network Protocols

Definition:

Network protocols are a set of rules that control how data is transmitted and received over a network.

Common Network Protocols

HTTP (HyperText Transfer Protocol):

It is a protocol used for transferring web pages between a web server and a browser.
It allows users to view websites on the internet.

HTTPS (HyperText Transfer Protocol Secure):

It is a secure version of HTTP that encrypts data for safe communication.
It protects sensitive information such as passwords and personal data.

FTP (File Transfer Protocol):

It is a protocol used to transfer files between computers over a network.
It allows uploading and downloading of files from a server.

SMTP (Simple Mail Transfer Protocol):

It is a protocol used for sending emails from one server to another.
It is mainly used for outgoing mail services.

POP3 (Post Office Protocol version 3):

It is a protocol used to receive emails by downloading them to a device.
Once downloaded, emails are usually removed from the server.

IMAP (Internet Message Access Protocol):

It is a protocol used to access emails directly from the mail server.
Emails remain on the server and can be accessed from multiple devices.

TCP (Transmission Control Protocol):

It is a protocol that ensures reliable and error-free data transmission.
It checks and delivers data in the correct order.

UDP (User Datagram Protocol):

It is a protocol that provides fast data transmission without error checking.
It is used where speed is more important than reliability.

IP (Internet Protocol):

It is a protocol responsible for identifying devices using IP addresses. It also helps in routing data to the correct destination.

DNS (Domain Name System):

It is a system that converts domain names into IP addresses. It helps users access websites using easy names instead of numbers.

DHCP (Dynamic Host Configuration Protocol):

It is a protocol that automatically assigns IP addresses to devices. It helps devices connect to a network without manual configuration.

17. IP Addressing**Introduction**

In a computer network, every device must have a unique identity so that data can be sent and received correctly. This unique identity is known as an IP address.

Definition of IP Address

An IP address is a unique numerical label assigned to each device connected to a computer network that is used to identify and communicate with that device.

Functions of IP Address

1. It identifies each device in a network uniquely.
2. It helps in sending and receiving data correctly.
3. It enables communication between different networks.
4. It helps in routing data across the internet.

Types of IP Address**17.1 IPv4 (Internet Protocol Version 4)****Definition**

IPv4 is a 32-bit addressing system used to identify devices in a network and enable communication between them.

Format of IPv4

IPv4 addresses are written in decimal form and divided into four parts separated by dots.

Example

192.168.1.1

Features of IPv4

1. It uses 32-bit addressing system.
2. It provides approximately 4.3 billion unique addresses.
3. It is widely used in current networking systems.
4. It is simple and easy to understand.

Limitations of IPv4

1. Limited number of addresses.
2. Increasing number of devices has caused address shortage.
3. Less efficient for modern networking requirements.

17.2 IPv6 (Internet Protocol Version 6)

Definition

IPv6 is a 128-bit addressing system developed to overcome the limitations of IPv4 and provide a large number of IP addresses.

Format of IPv6

IPv6 addresses are written in hexadecimal form and separated by colons.

Example

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Features of IPv6

1. It uses 128-bit addressing system.
2. It provides a very large number of IP addresses.
3. It improves security and performance.
4. It supports modern internet devices and technologies.

Advantages of IPv6

- Unlimited address availability
- Better security features
- Faster data processing
- Efficient routing system

Difference between IPv4 and IPv6

IPv4	IPv6
Uses a 32-bit IP address.	Uses a 128-bit IP address.
Uses decimal dot-separated notation (e.g., 192.168.0.1).	Uses hexadecimal colon-separated notation (e.g., 2001:db8::1).
Provides a limited address space of about 4.3 billion addresses.	Provides an extremely large address space for future growth.
Supports manual configuration and DHCP.	Supports SLAAC, DHCPv6, and manual configuration.
End-to-end connectivity is often affected due to NAT.	End-to-end connectivity is restored without NAT.
IPsec support is optional.	IPsec support is built into the protocol design.
Fragmentation is performed by both sender and routers.	Fragmentation is performed only by the sender.
Does not support flow-based packet identification.	Uses a Flow Label field for packet flow identification.
Includes a header checksum.	Does not include a header checksum.
Supports broadcast communication.	Uses multicast and anycast instead of broadcast.
Header size is variable (20–60 bytes).	Header size is fixed at 40 bytes.
Uses address classes (A, B, C, D, and E).	Does not use address classes.
Supports Variable Length Subnet Masking (VLSM).	Uses prefix-based addressing.
Example address: 66.94.29.13.	Example address: 2001:db8::1.

6. Default Gateway

Definition:

A default gateway is a network device that connects a local network to other networks, usually the internet.

Explanation:

When a computer wants to send data outside its own network, it sends the data to the default gateway. The gateway then forwards the data to the correct destination on other networks.

Example:

In a home network, the router acts as the default gateway.

Roles of Default Gateway:

- Connects the local network to the internet
- Routes data from the local network to external networks
- Acts as an access point for communication outside the network

How to Check Default Gateway in CMD

Steps:

1. Open Command Prompt
2. Type: ipconfig
3. Press Enter

You will see a line called Default Gateway showing the gateway address.

7. Network Command Line Utilities

7.1. Ping

Definition:

Ping is a command used to check whether a network connection to another device or website is working.

How to run in CMD:

1. Open Command Prompt
2. Type: ping google.com
3. Press Enter

What it does:

- Sends small data packets to the destination
- Shows if the connection is successful and how fast it is

7.2. IPConfig

Definition:

IPConfig is a command used to display the IP address and network details of a computer.

How to run in CMD:

1. Open Command Prompt
2. Type: ipconfig
3. Press Enter

What it shows:

- IP address
- Subnet mask
- Default gateway

7.3. Tracert

Definition:

Tracert is a command used to trace the path taken by data from the source to the destination.

How to run in CMD:

1. Open Command Prompt
2. Type: tracert google.com
3. Press Enter

What it does:

- Shows each step (router) the data passes through
- Helps find where delays or problems occur

7.4. NSLookup

Definition:

NSLookup is a command used to find the IP address of a domain name.

How to run in CMD:

1. Open Command Prompt
2. Type: nslookup google.com
3. Press Enter

What it does:

- Converts a website name into its IP address
- Helps in troubleshooting DNS problems

9. Internet, Intranet, and Extranet

19.1 Internet

Definition

The Internet is a global network of interconnected computers that allows users to access and share information worldwide.

Features of Internet

1. It is a public network accessible to everyone.
2. It connects millions of computers globally.

3. It provides services like email, web browsing, and file sharing.
4. It is not owned by any single organization.

Uses of Internet

- Online communication
- Education and research
- E-commerce
- Entertainment

Advantages of Internet

- Global communication
- Easy access to information
- Online services availability

Disadvantages of Internet

- Security threats
- Addiction and misuse
- Spread of false information

19.2 Intranet

Definition

An intranet is a private network used within an organization to share information, resources, and services among authorized users.

Features of Intranet

1. It is a private network.
2. It is accessible only within an organization.
3. It provides secure communication.
4. It uses internet technologies internally.

Uses of Intranet

- Office communication
- Employee information sharing
- Internal document management

Advantages of Intranet

- High security

- Fast communication
- Controlled access

Disadvantages of Intranet

- Limited access
- Expensive maintenance

19.3 Extranet

Definition

An extranet is a private network that allows controlled access to external users such as clients, partners, and suppliers.

Features of Extranet

1. It is a partially private network.
2. It allows external authorized users.
3. It improves business collaboration.
4. It is more secure than the internet.

Uses of Extranet

- Business communication with partners
- Online order tracking systems
- Supplier management systems

Advantages of Extranet

- Improves business collaboration
- Secure data sharing
- Efficient communication with external users

Disadvantages of Extranet

- Security risks if not properly managed
- High setup and maintenance cost

Difference between Internet, Intranet and Extranet

Point of Difference	Internet	Intranet	Extranet
Accessibility	Public – open to all	Private – only employees	Private – employees + authorized external users

Coverage	Worldwide	Within an organization	Organization + selected partners/clients
Purpose	Global information sharing	Internal communication & collaboration	Secure collaboration with external stakeholders
Security	User - dependent; less controlled	Protected by organizational firewalls	Firewalls + encryption + controlled login
Ownership	No single owner	Single organization	Single or multiple organizations
Users	General public	Employees of the organization	Employees + suppliers/partners/clients
Maintenance	Maintained by ISPs and global organizations	Maintained by the organization's IT team	Maintained jointly by the organization and external partners
Example	Public web browsing	Wipro's internal HR portal	Dell–Intel supplier network